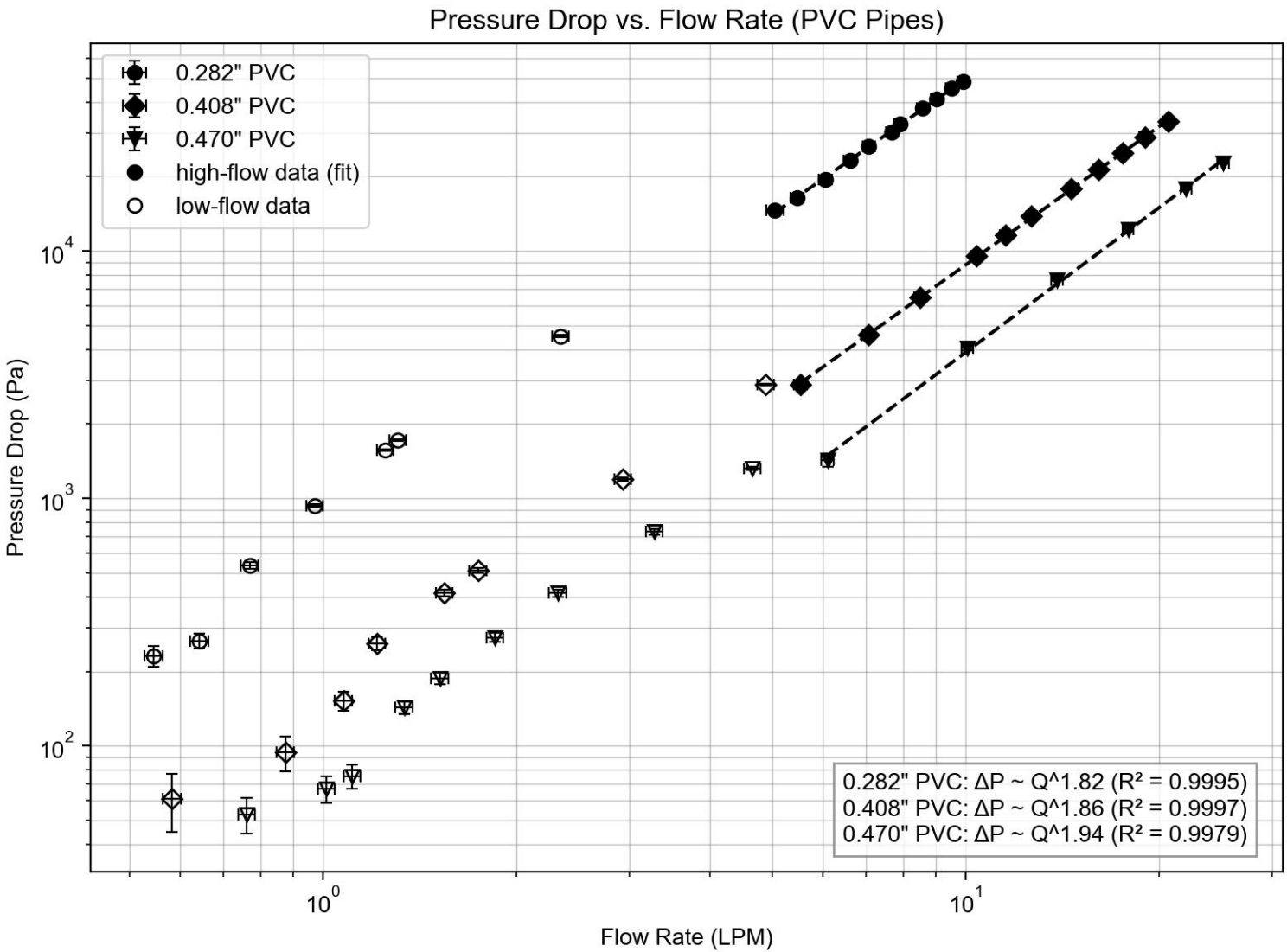
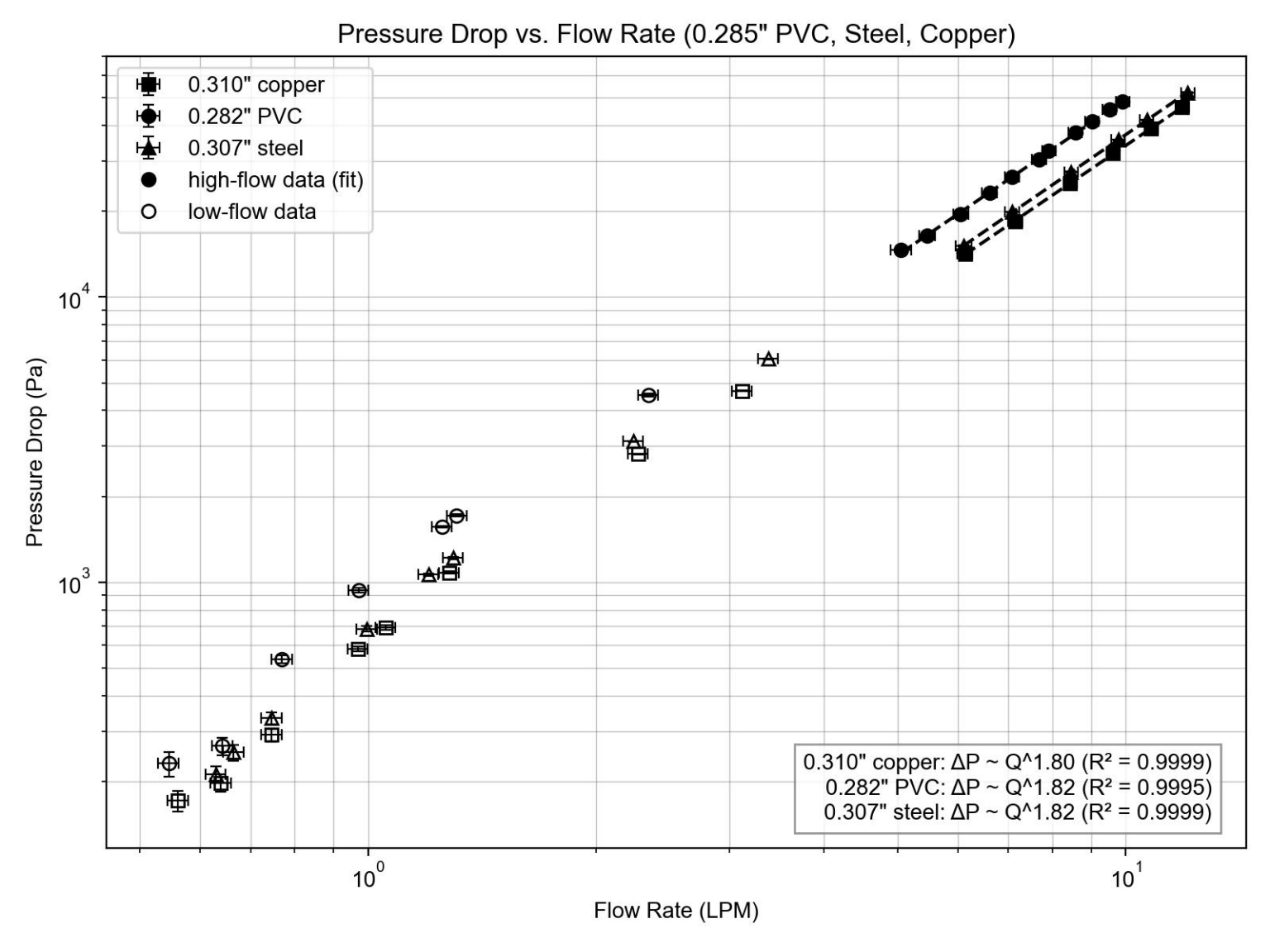
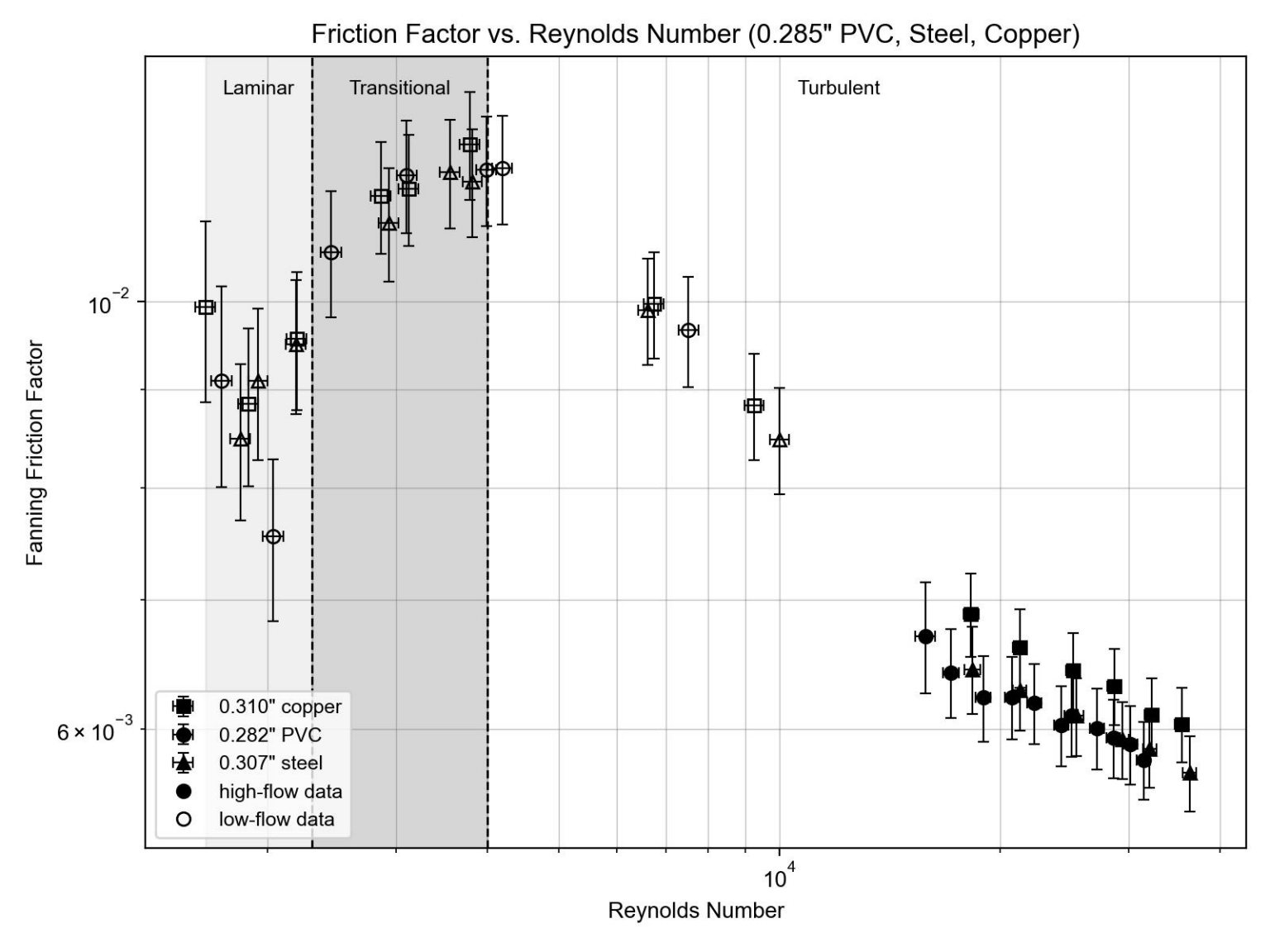
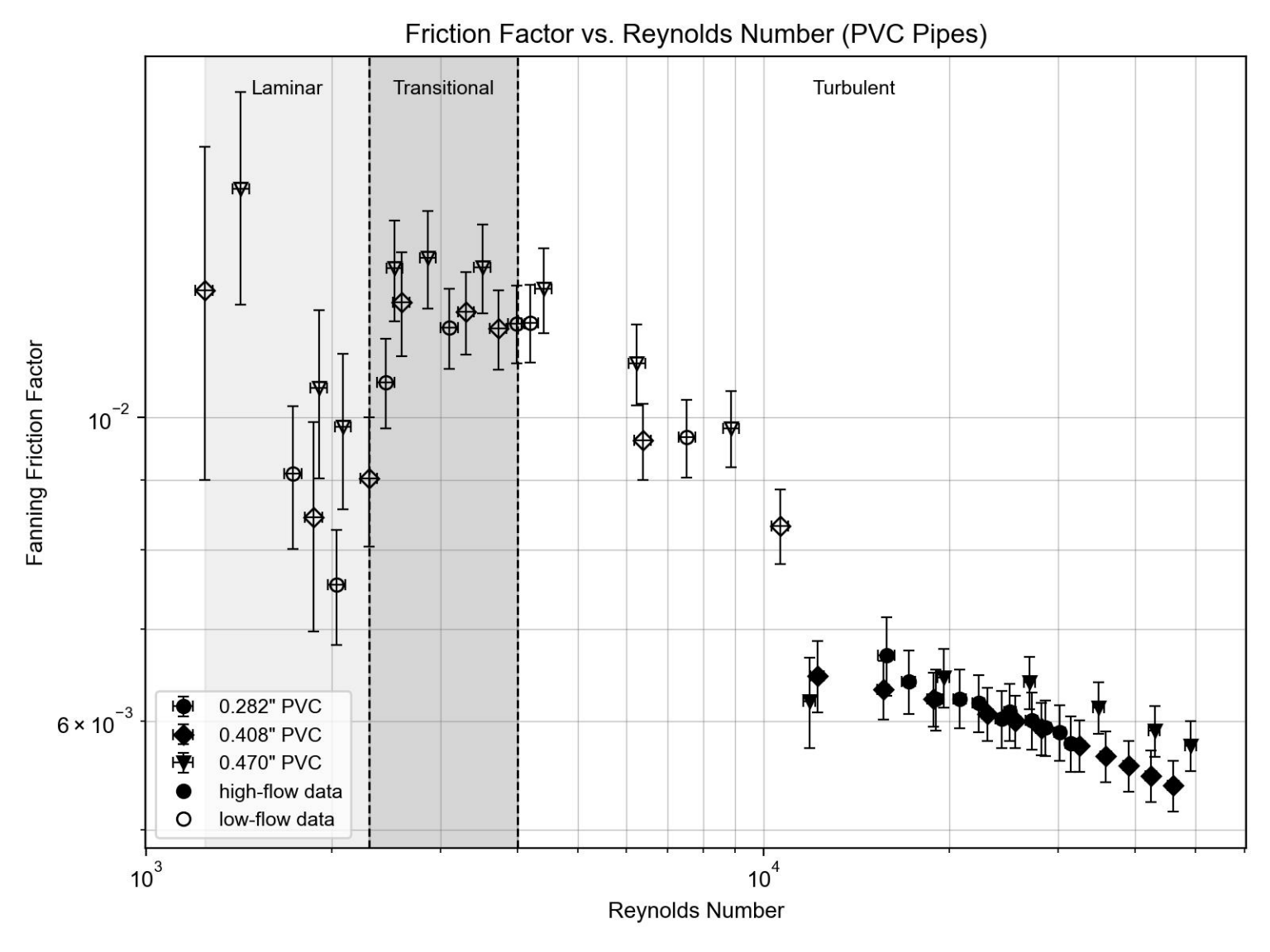
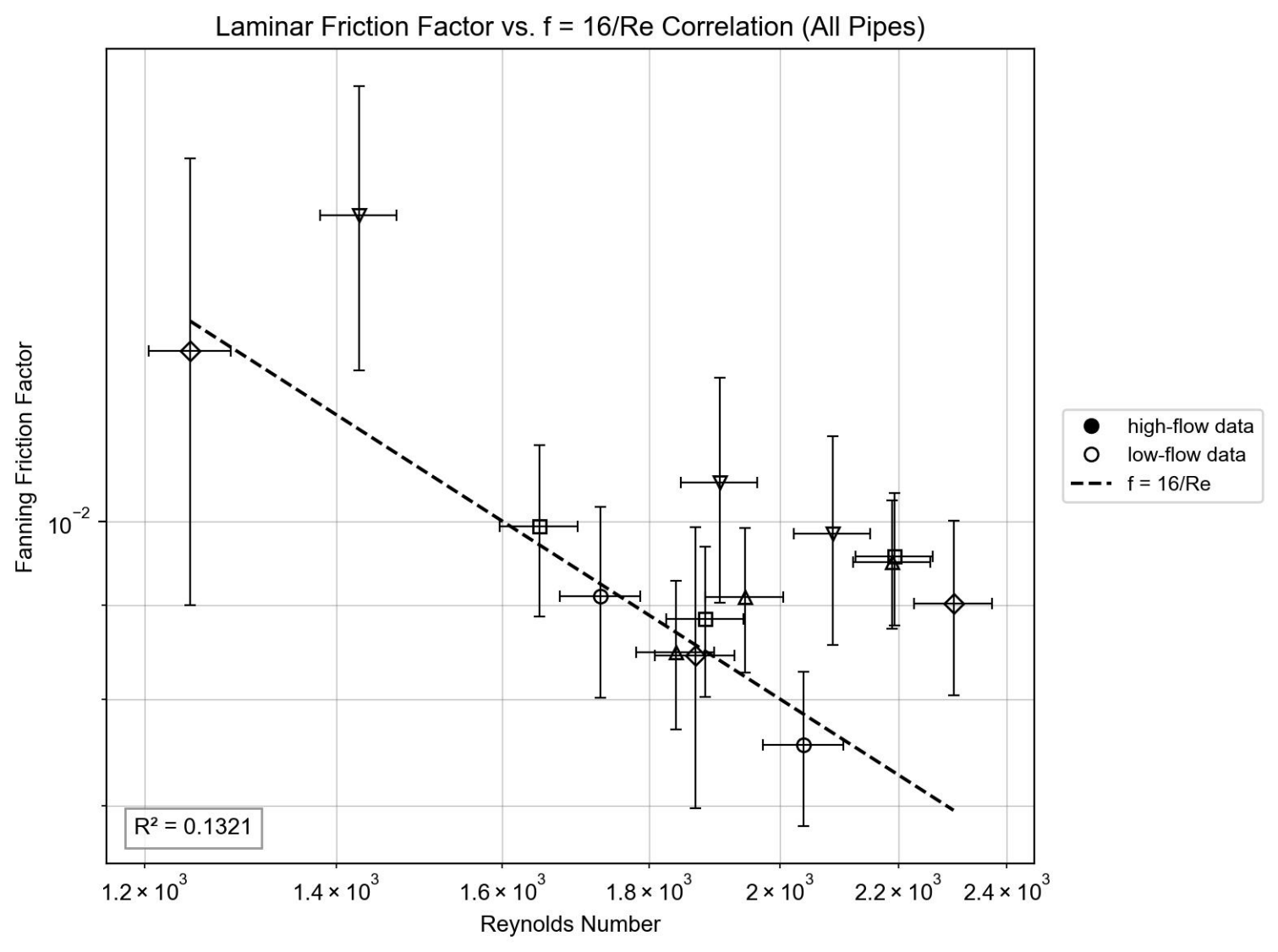
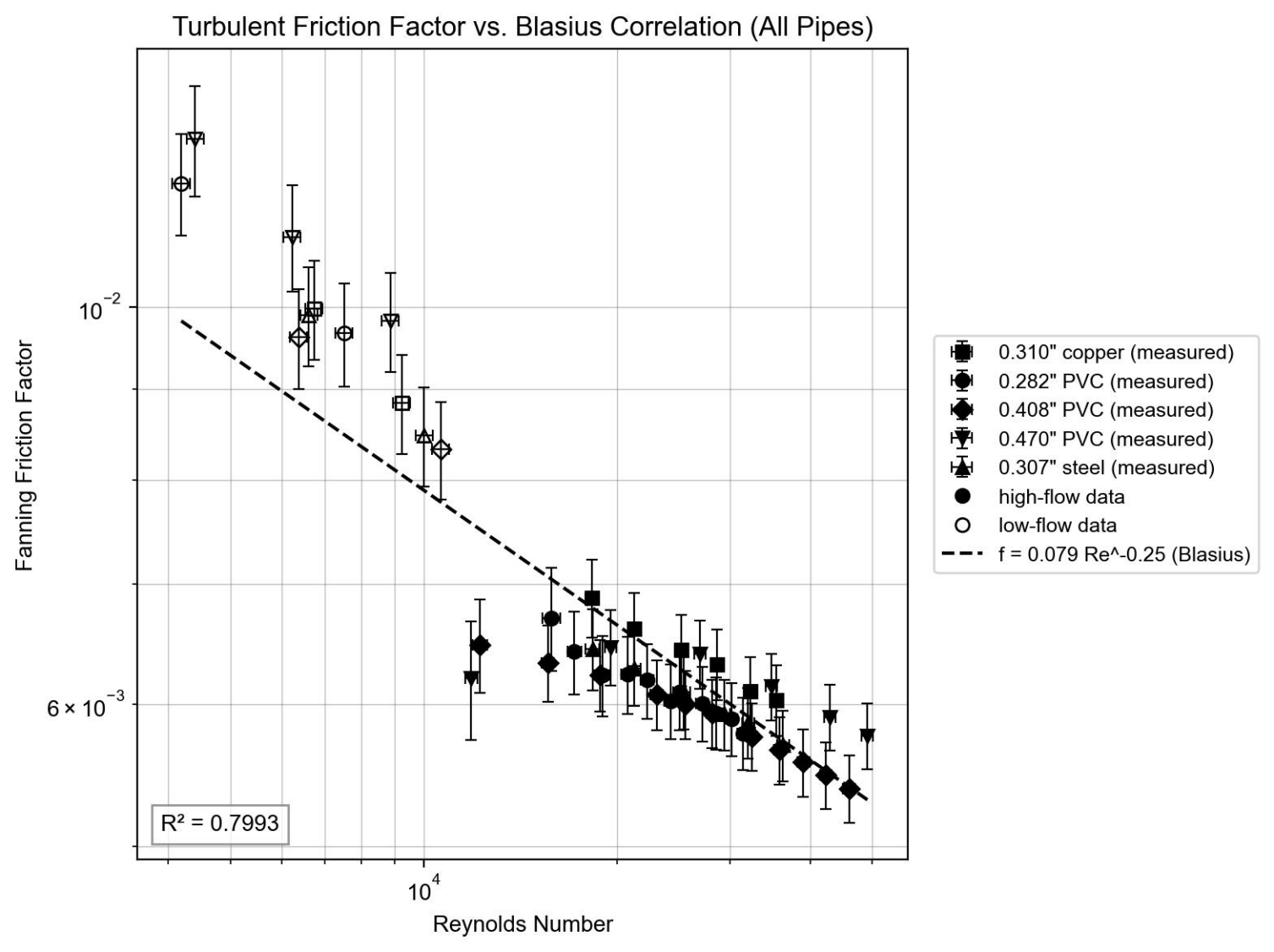

Pressure Drop vs. Flow Rate (PVC Pipes)
0.282" PVC
0.408" PVC
0.470" PVC
high-flow data (fit)
low-flow data
Pressure Drop (Pa)
Flow Rate (LPM)
0.282" PVC: ΔP ~ Q^1.82 (R² = 0.9995)
0.408" PVC: ΔP ~ Q^1.86 (R² = 0.9997)
0.470" PVC: ΔP ~ Q^1.94 (R² = 0.9979)

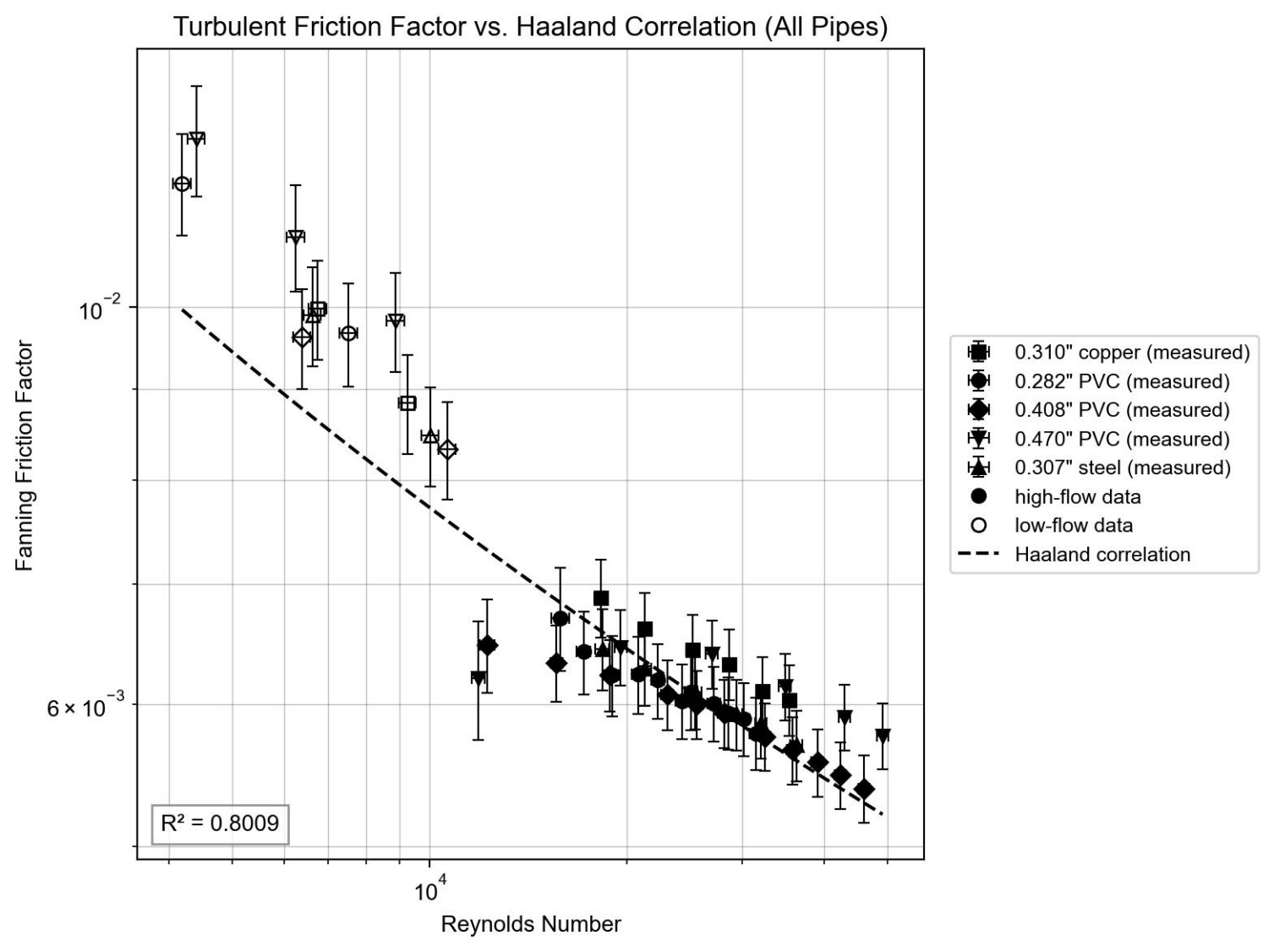

Pressure Drop vs. Flow Rate (0.285" PVC, Steel, Copper)
0.310" copper
0.282" PVC
0.307" steel
high-flow data (fit)
low-flow data
Pressure Drop (Pa)
Flow Rate (LPM)
10^4
10^3
10^0
10^1
0.310" copper: ΔP ~ Q^1.80 (R² = 0.9999)
0.282" PVC: ΔP ~ Q^1.82 (R² = 0.9995)
0.307" steel: ΔP ~ Q^1.82 (R² = 0.9999)

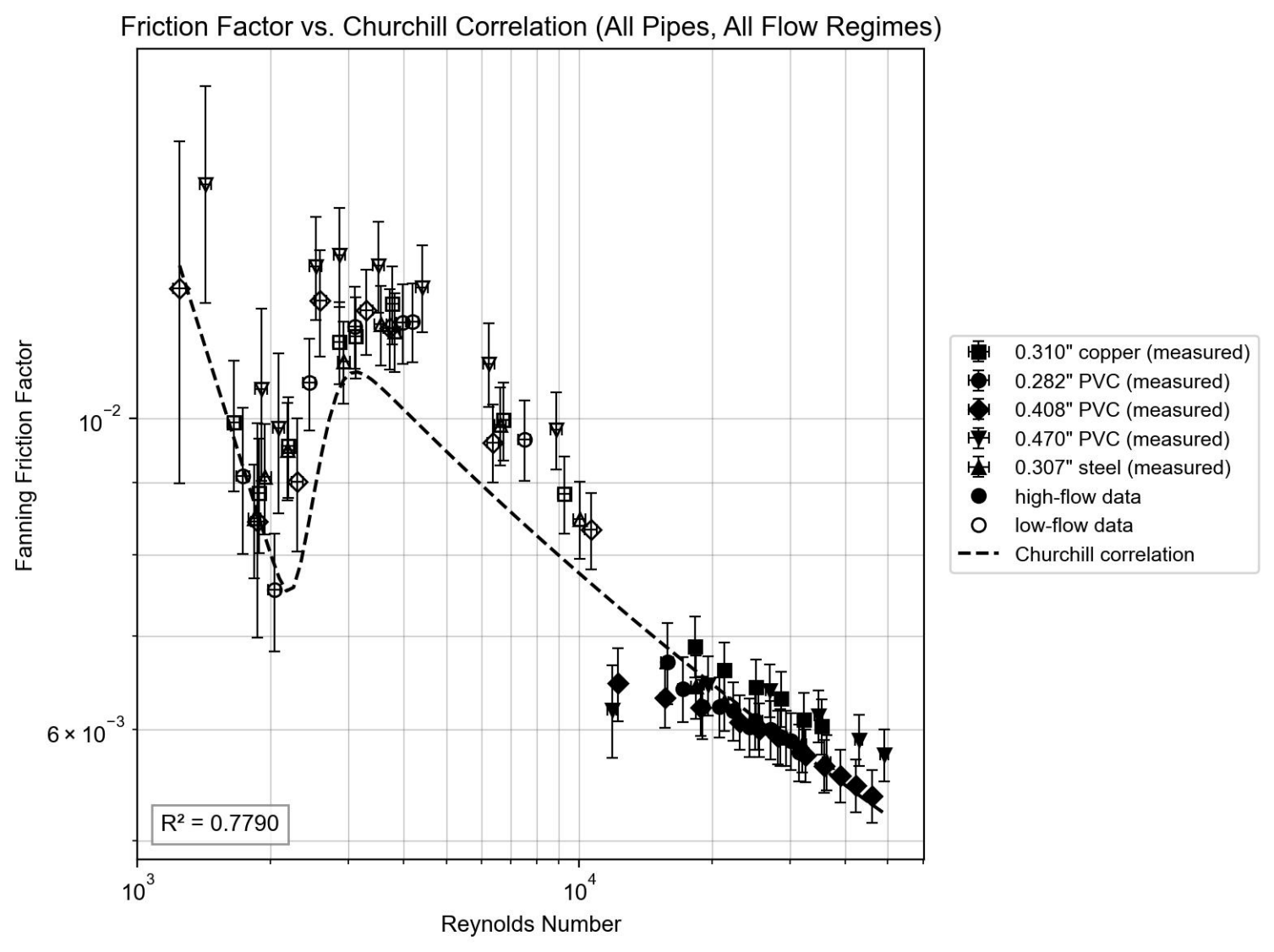

Friction Factor vs. Reynolds Number (0.285" PVC, Steel, Copper)
Laminar
Transitional
Turbulent
10^{-2}
6×10^{-3}
Fanning Friction Factor
10^{4}
Reynolds Number
0.310" copper
0.282" PVC
0.307" steel
high-flow data
low-flow data

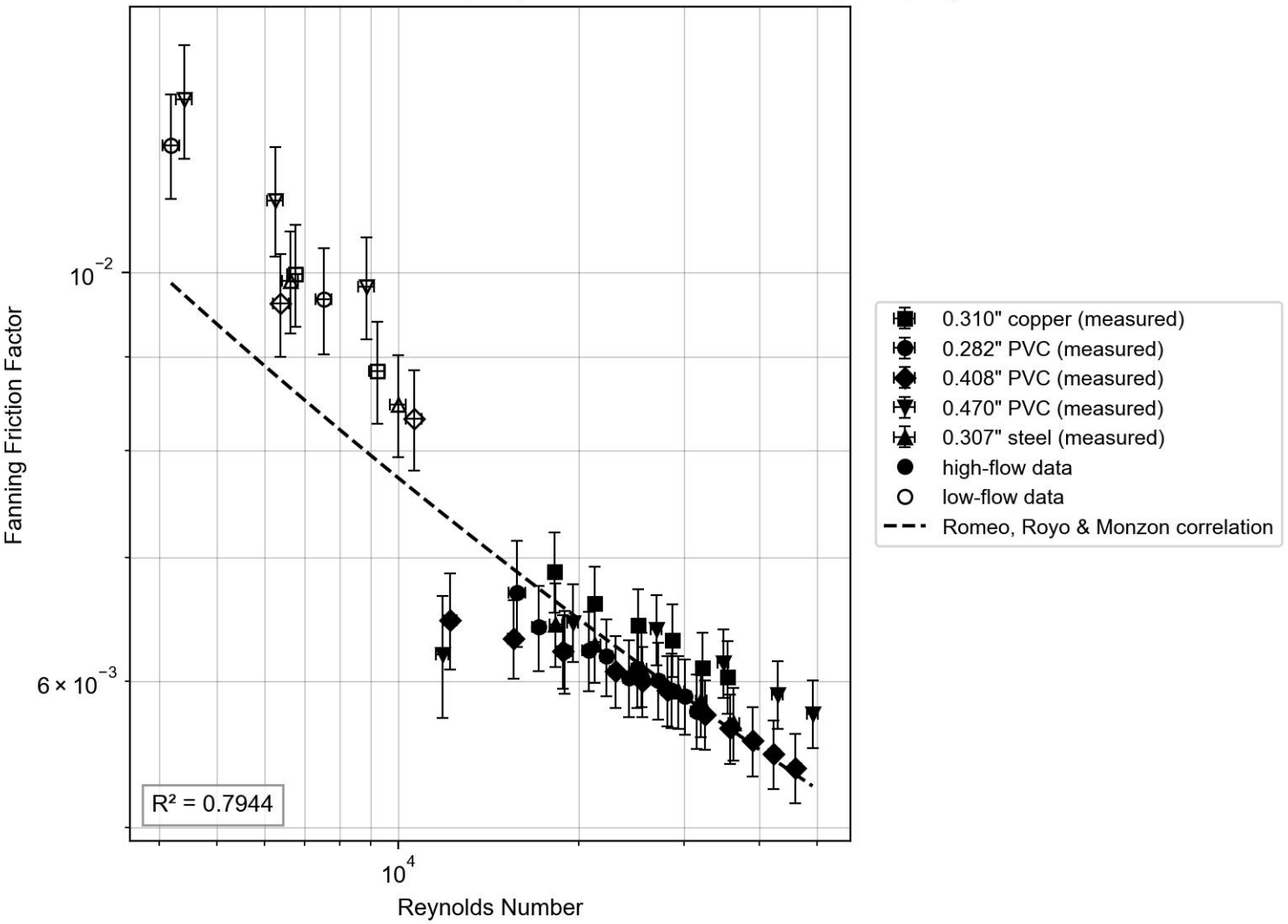

Friction Factor vs. Reynolds Number (PVC Pipes)
Laminar
Transitional
Turbulent
Fanning Friction Factor
10^{-2}
6×10^{-3}
10^{3}
10^{4}
Reynolds Number
0.282" PVC
0.408" PVC
0.470" PVC
high-flow data
low-flow data

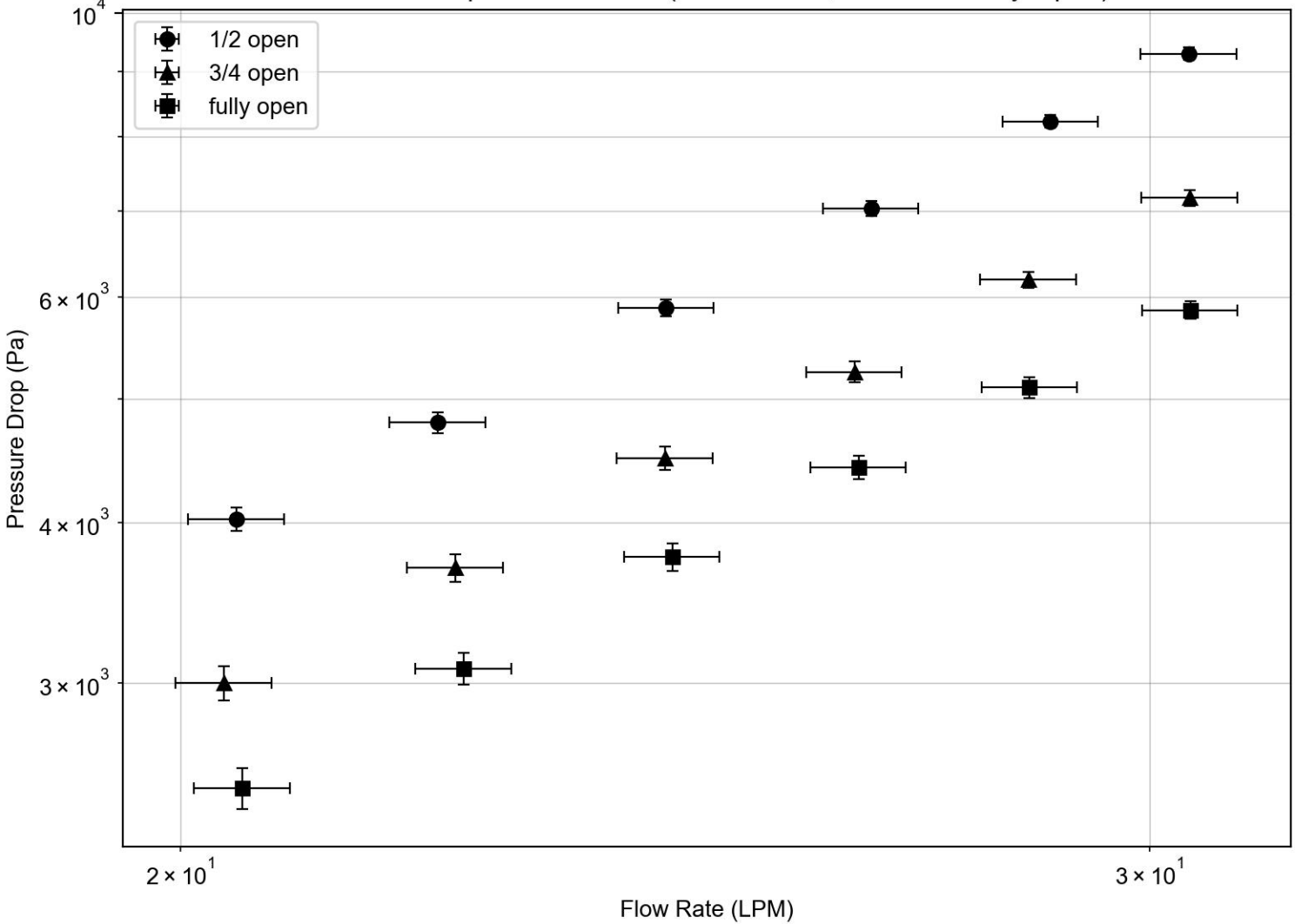

Laminar Friction Factor vs. f = 16/Re Correlation (All Pipes)
Fanning Friction Factor
10^{-2}
$R^2 = 0.1321$
1.2×10^3
1.4×10^3
1.6×10^3
1.8×10^3
2×10^3
2.2×10^3
2.4×10^3
Reynolds Number
high-flow data
low-flow data
f = 16/Re

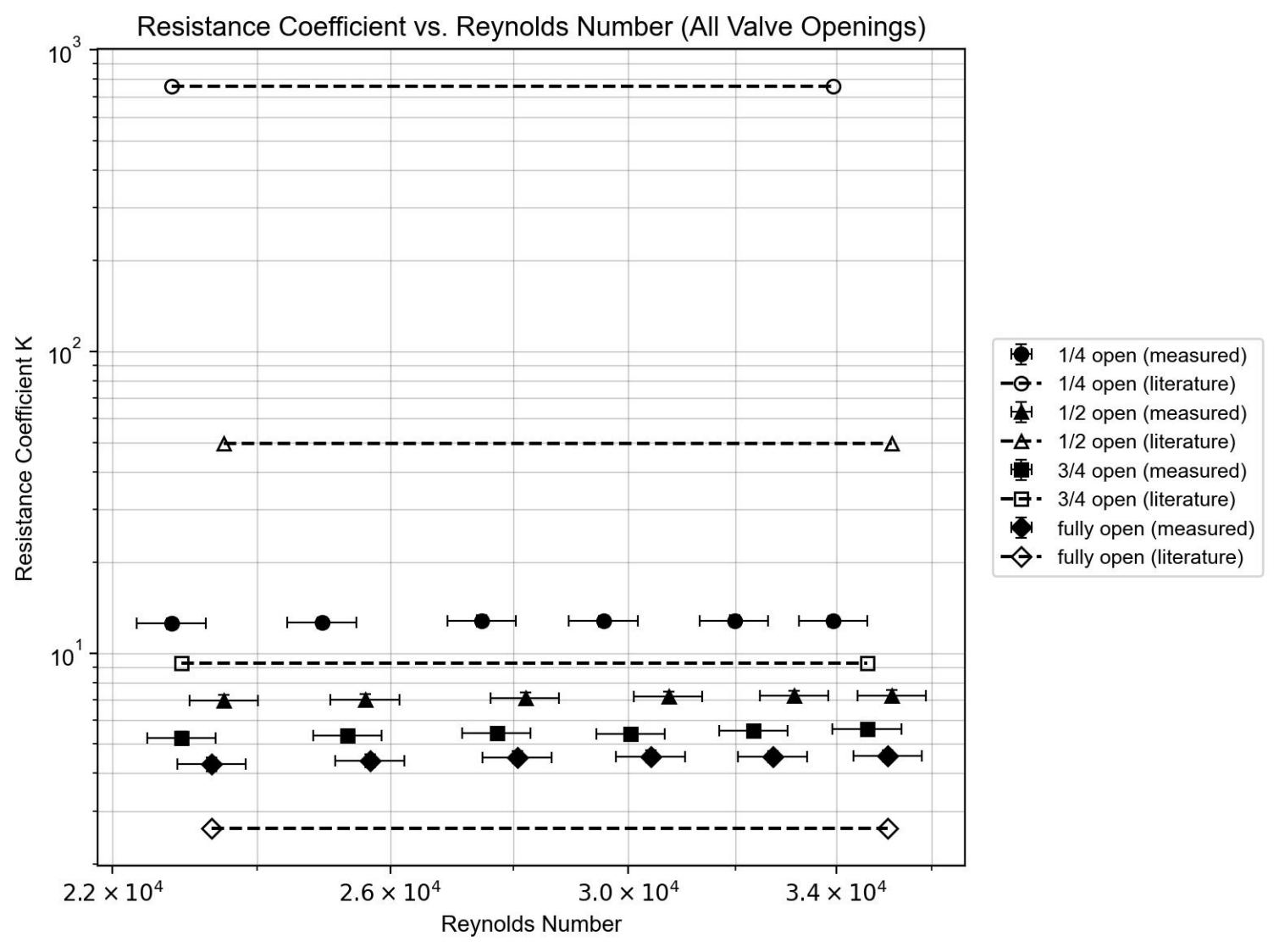

Turbulent Friction Factor vs. Blasius Correlation (All Pipes)
Fanning Friction Factor
Reynolds Number
10^{-2}
6×10^{-3}
10^{4}
R² = 0.7993
0.310" copper (measured)
0.282" PVC (measured)
0.408" PVC (measured)
0.470" PVC (measured)
0.307" steel (measured)
high-flow data
low-flow data
f = 0.079 Re^-0.25 (Blasius)

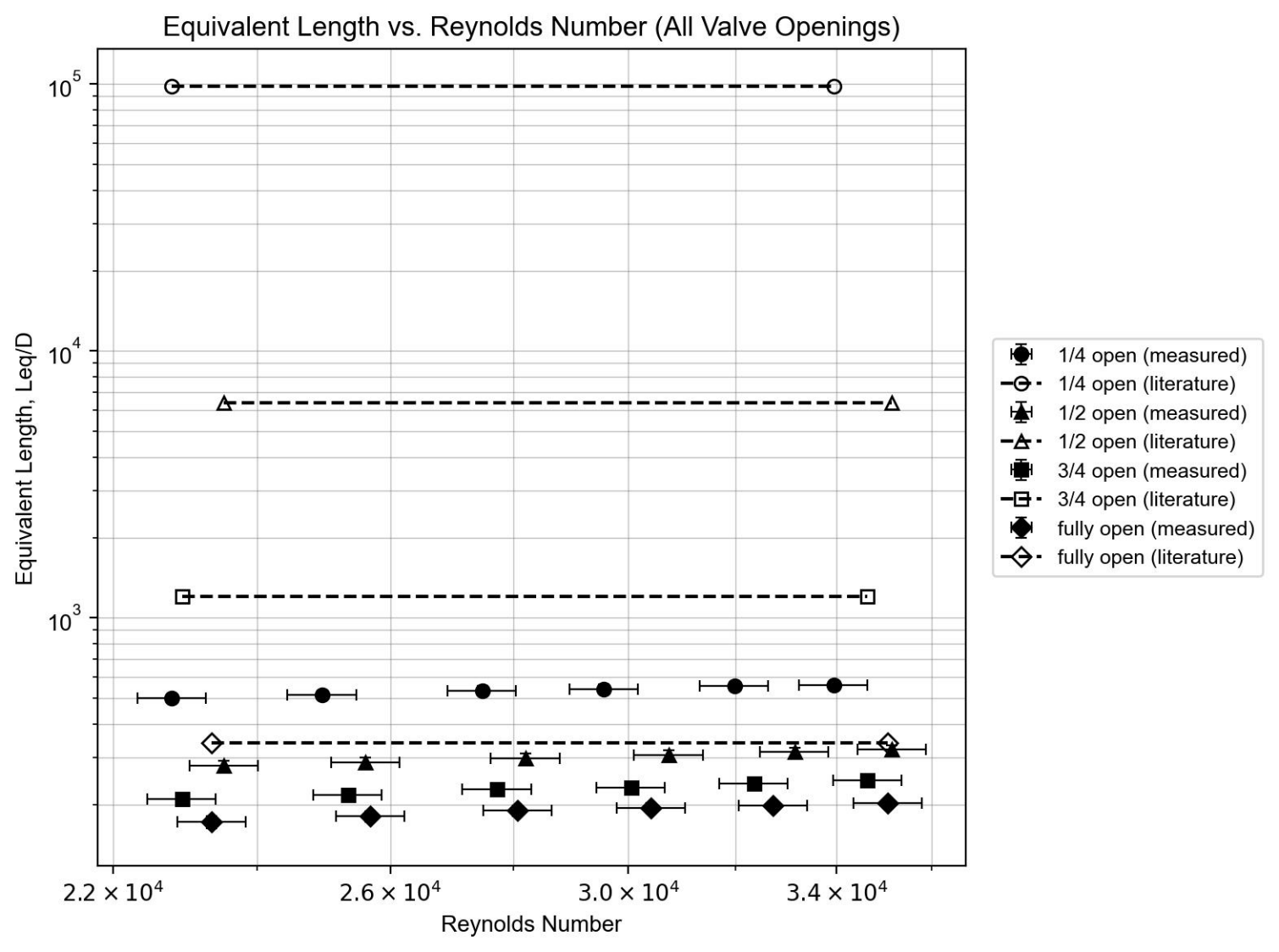

Turbulent Friction Factor vs. Haaland Correlation (All Pipes)
Fanning Friction Factor
Reynolds Number
10^{-2}
6×10^{-3}
10^{4}
R² = 0.8009
0.310" copper (measured)
0.282" PVC (measured)
0.408" PVC (measured)
0.470" PVC (measured)
0.307" steel (measured)
high-flow data
low-flow data
Haaland correlation

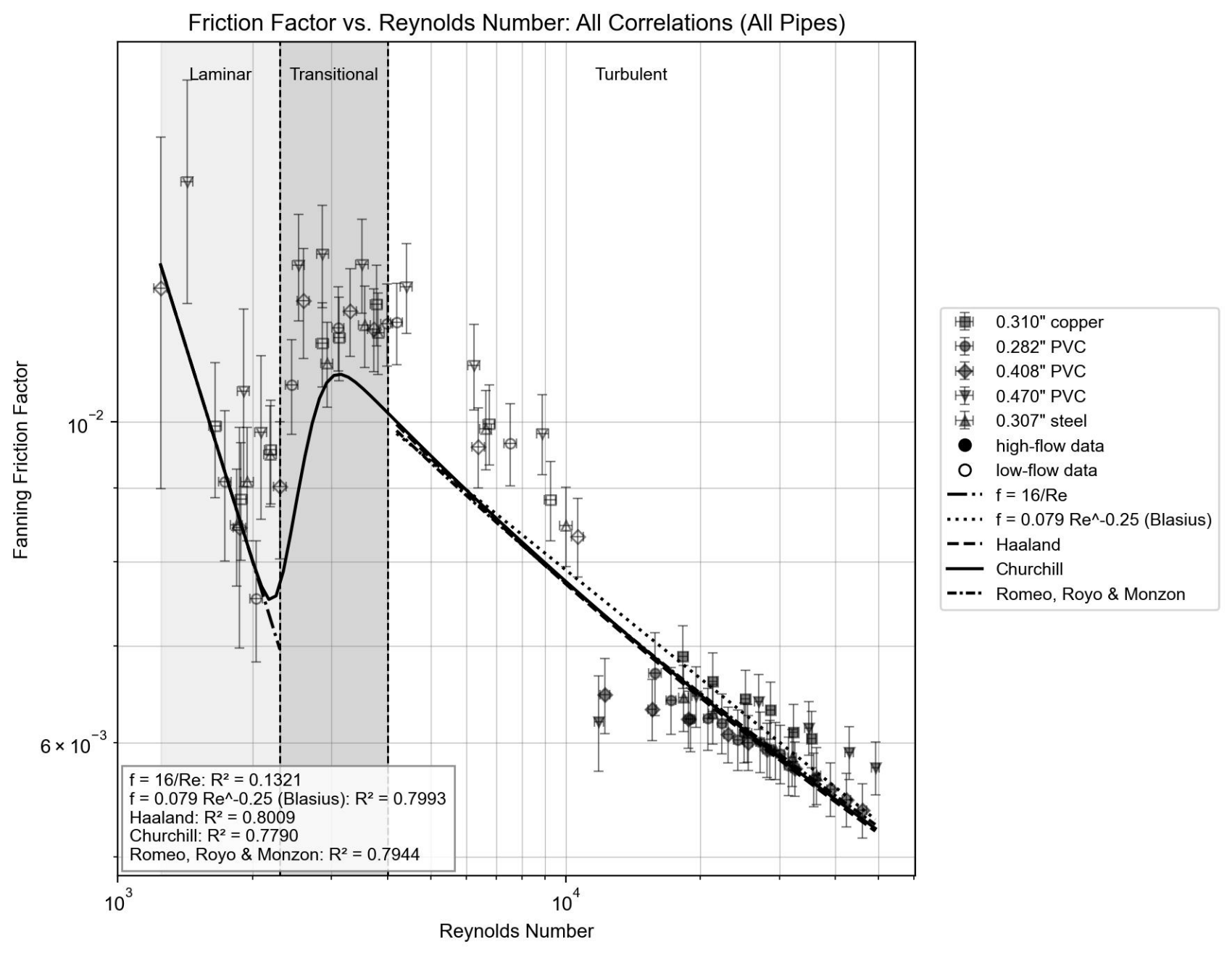

Friction Factor vs. Churchill Correlation (All Pipes, All Flow Regimes)
Fanning Friction Factor
Reynolds Number
10^{-2}
6×10^{-3}
10^{3}
10^{4}
R² = 0.7790
0.310" copper (measured)
0.282" PVC (measured)
0.408" PVC (measured)
0.470" PVC (measured)
0.307" steel (measured)
high-flow data
low-flow data
Churchill correlation


Turbulent Friction Factor vs. Romeo, Royo & Monzon Correlation (All Pipes)
Fanning Friction Factor
10^{-2}
6×10^{-3}
10^{4}
Reynolds Number
R² = 0.7944
0.310" copper (measured)
0.282" PVC (measured)
0.408" PVC (measured)
0.470" PVC (measured)
0.307" steel (measured)
high-flow data
low-flow data
Romeo, Royo & Monzon correlation


Pressure Drop vs. Flow Rate (Globe Valve, 1/2 / 3/4 / Fully Open)
1/2 open
3/4 open
fully open
10^4
6×10^3
4×10^3
3×10^3
Pressure Drop (Pa)
2×10^1
3×10^1
Flow Rate (LPM)


Resistance Coefficient vs. Reynolds Number (All Valve Openings)
Resistance Coefficient K
10^3
10^2
10^1
2.2×10^4
2.6×10^4
3.0×10^4
3.4×10^4
Reynolds Number
1/4 open (measured)
1/4 open (literature)
1/2 open (measured)
1/2 open (literature)
3/4 open (measured)
3/4 open (literature)
fully open (measured)
fully open (literature)


Equivalent Length vs. Reynolds Number (All Valve Openings)
Equivalent Length, Leq/D
10^5
10^4
10^3
2.2×10^4
2.6×10^4
3.0×10^4
3.4×10^4
Reynolds Number
1/4 open (measured)
1/4 open (literature)
1/2 open (measured)
1/2 open (literature)
3/4 open (measured)
3/4 open (literature)
fully open (measured)
fully open (literature)


Friction Factor vs. Reynolds Number: All Correlations (All Pipes)
Laminar
Transitional
Turbulent
Fanning Friction Factor
Reynolds Number
10^{-2}
6×10^{-3}
10^{3}
10^{4}
f = 16/Re: R² = 0.1321
f = 0.079 Re^-0.25 (Blasius): R² = 0.7993
Haaland: R² = 0.8009
Churchill: R² = 0.7790
Romeo, Royo & Monzon: R² = 0.7944
0.310" copper
0.282" PVC
0.408" PVC
0.470" PVC
0.307" steel
high-flow data
low-flow data
f = 16/Re
f = 0.079 Re^-0.25 (Blasius)
Haaland
Churchill
Romeo, Royo & Monzon